Optik 126 (2015) 3771–3780

Contents lists available at ScienceDirect

Optik

journal homepage: www.elsevier.de/ijleo

One-shot monochromatic symbol pattern for 3D reconstruction using perfect submap coding

Mei Fang[a,b], Wei Shen[a], Dan Zeng[a], Xiaojun Jia[b], Zhijiang Zhang[a,*]

[a] Shanghai University Key Laboratory of Specialty Fiber Optics and Optical Access Networks, Shanghai 200072, China
[b] Jiaxing University College of Mathematics Physics and Information Engineering, Zhejiang 314001, China

ARTICLE INFO

Article history:
Received 30 June 2014
Accepted 22 July 2015

Keywords:
Machine vision
Coded structured light
Monochromatic symbol pattern
Perfect submap

ABSTRACT

Monochromatic perfect submap coding is a technique which utilizes structured light to reconstruct three-dimensional shapes. This coding strategy is robust against occlusions, suitable for dynamic scene detection, and less sensitive to colorful surface when compared to color coding. However, perfect submap patterns are often poor in terms of resolution and accuracy. In this work, the symbol density spectrum (SDS) and its features are used to choose symbols for improving resolution and decreasing decoding error of perfect submap symbol patterns. The proposed SDS provides a distribution of symbol points for reconstruction after which 10 symbol features are extracted. Following a comparative analysis of the symbol features and scene testing of symbols damage rate, a new density pattern is formed using nine symbols chosen from one group. A 3D reconstruction experiment showed that results obtained by this method are of a higher resolution compared with other existing monochromatic spatial coding techniques and are robust against color surface.

© 2015 Elsevier GmbH. All rights reserved.

1. Introduction

Structured light used in 3D shape measurements provides several advantages including non-contact, flexibility, easy control and high contrast [1,2]. One-shot or multiple patterns are projected onto the target object using a projector, and the coded information produced on the surface is captured by a charge-coupled device (CCD) camera in order to recover a 3D image.

Generally, multiple patterns achieve higher spatial resolution and accuracy, however multiple patterns are not the optimal choice for detecting objects in a dynamic scene due to hardware limitations. In order to build a real-time system, the numbers of projection should be less than 7 if the speed of projection is 180 Hz. The maximum projection speed of the current digital display video projector is 120 frames/s [3]. To reach high-speed 3D shape measurement, the structured patterns must be switched rapidly, and captured in a short period of time. It requires that the projector and the camera must be in focus to alleviate errors induced by the image blurring or defocus [3,4].

The one-shot pattern offers advantages in dynamic scenes. For example, discrete spatial multiplexing coded by De Bruijn can accurately reconstruct moving objects [2]. Compared with multiple patterns, however, shape reconstruction using one-shot pattern coding is often weaker in resolution and accuracy. In addition, colors coding in discrete spatial multiplexing may be easily influenced by surface texture.

To address these problems, we follow the one-shot pattern approach to enhance the spatial resolution and robustness of 3D reconstruction. Special symbols were developed to replace the primitives in the perfect submap (PSM). In the current approach, a monochromatic coding method which is robust against colored objects is adopted. Then, compared with De Bruijn, PSM is coded by both axes which allows for possible recovery from neighbors in near regions of discontinuity. Finally, symbol features are extracted in density spectrum (SDS) and symbol. The key points of symbols with low or high density can be distinguished through symbol features comparison and scene testing. The set of symbols with the optimal properties of easy decoding and high resolution are selected.

The rest paper is organized as follows. Section 2 introduces the related work. Section 3 introduces the construction of PSM applicable to the system dealt with in the present paper. In Section 4, our proposed symbol pattern is introduced. In Section 5, experiments and analysis are conducted. Conclusion is drawn in the last section.

* Corresponding author. Tel.: +86 21 56332164.
E-mail address: zjzhang@shu.edu.cn (Z. Zhang).

http://dx.doi.org/10.1016/j.ijleo.2015.07.140
0030-4026/© 2015 Elsevier GmbH. All rights reserved.

Table 1
Typical one-shot coding pattern by discrete coding spatial multiplexing.

Technique	Color surface
1. De Bruijn (or related conception has one-dimension)	
1.1 Color stripe [14,15]	[14] Y [15] N
1.2 Monochromatic [31]	Y
2. M-array (or related conception has two-dimension)	
2.1 Color [3,16–20]	N
2.2 Monochromatic [22–27]	Y

2. Related work

2.1. One-shot shape acquisition

The coded pattern is the most critical factor in improving resolution, accuracy and precision of 3D reconstruction, and also determines whether the method is suitable for dynamic scene detection. A one-shot coding strategy is possible for real-time 3D shape detection, which can prove useful for applications such as one-shot phase shifting and discrete coding spatial multiplexing [2,5].

One-shot phase shifting is comprised of single or multiple phase shifting and generates a pattern using a gray or color image (e.g. color stripe pattern). A new projection device (e.g. DLP) modulates a single or multiple fringe pattern into a single gray [6,7] or color image [8,9]. However, one-shot multiple fringe pattern cannot be applied in high accuracy measurements, as the image produced by two or more patterns has a small range of gray levels and it is poor in phase resolution [10,11]. Methods based on phase calculation projection techniques for single-shot or real-time 3D shape measurement, have been described previously [3,10].

Discrete coding spatial multiplexing, which is usually coded by De Bruijn, M-array, and PSM, provides a approach to one-shot patterns. Using the De Bruijn sequence, Sagawa et al. proposed a method to densely reconstruct moving objects which only utilizes two colors of De Bruijn sequence and does not require explicit correspondences between a camera and a projector [2]. A pattern coded with vertical slits using a De Bruijn sequence can obtain sub-pixel accuracy [12,13].

De Bruijn coding adopts a single axis which can cause difficulty in decoding a modulated pattern due to occlusions and object shadows related to ambiguities in the decoding step [1]. Techniques that coding both axes (e.g., M-array, perfect submap) can help recover 3D surface from neighbors in regions of discontinuity [3,10].

2.2. Monochromatic and color pattern for discrete coding by spatial multiplexing

Among discrete coding spatial multiplexing, the primitive in pattern can be coded by color stripe [14,15], spot [16–19], diamond [19–21], monochromatic geometric features [22–27], or hybrid coding [13,28].

Most discrete coding spatial multiplexing is coded by monochromatic or color patterns. As shown in Table 1, monochromatic coding is less sensitive to color surfaces compared with color coding. In the coding pattern, a smaller window makes codes position information efficiently. However, a small window causes an increase in the number of colors or monochromatic symbols in the pattern, according to M-array formula. As more colors are coded in the pattern, the shape reconstruction is more easily influenced by the texture of the surface, which tends to cause instability in results. Zhang et al. [13] proposed a 125-slit pattern with eight colors with a window size of three which can achieve sub-pixel accuracy for reconstruction. Pages et al. [15] used four colors to produce a 128-strip pattern which improved robustness. In order to reduce color coding influence, Zhang et al. [14] proposed regularized color coded by De Bruijn sequence, which divided the Euclidean norm of the RGB color by the relevant colors. Comparative experiments show that this method is better than other RGB color coding methods with regards to the treatment of dull surfaces, surface orientation, and illumination direction. The limitations of color coding have been discussed previously [1,13,29,30].

Table 2
Related concepts about M-array. A matrix **P** is a C-ary(r, s ; m, n)

Coding	Explanation
M-array	A $m \times n$ array as a sub-matrix in an $r \times s$ ($rs = 2^{mn}$)array appears only once [1].
De Bruijn	Special case when $r = m = 1$ (or $s = n = 1$) in M-array [33].
m-array	As M-array, but $rs = 2^{mn} - 1$, except zero matrix [33].
Pseudorandom arrays	Same as m-array [34,35].
Pseudorandom sequence	As De Bruijn, but not contain the substring formed by 0's [36].
Perfect maps	Same as M-array [34].
Perfect submap	As M-array, since not all the $m \times n$ submatrices are used [12].
Subperfect submaps	As Perfect submap, since the unicity of codes is not assuring, it is restrained by epipolar lines [26].

Unlike the primitive in color coding mentioned above, the primitive in monochromatic pattern coded by M-array, PSM, and other methods (Table 2) is often replaced by different symbols, such as disk, ring, segment [23], square, cuneiform [25,26], thickened cuneiform [27], or specially designed symbols [24,32]. Albitar et al. [23] used three different symbols to replace primitives in a 27×29 pattern, with a 3×3 symbol window appearing only once in the whole pattern. Since the pattern is coded by monochromatic symbols, the solution was robust against color objects. An M-array coding strategy is robust against occlusions(up to certain limit), however, nearly 783 points are gained during detection by this one-shot pattern method. Reiss et al. [27] improved resolution by using five different symbols, each of which contained four or six points for symbol reconstruction. Maurice et al. [25,26] proposed the largest subperfect submap pattern to date, with a pattern size of 100×150. The pattern gained 15,000 points for reconstruction since each symbol has only one key point. Jia et al. [24] proposed special symbols in a M-array pattern which gained more points(27,966) for reconstruction than other methods coded by monochromatic M-array, PSM, and other methods (Table 2). There are six points for each symbol in the 79×59 pattern. However, such symbols pattern unable to measure the irregular surface due to the key points adhering together when decoding a modulate pattern.

In order to improve resolution, the number of key points in a symbol need to be increased for reconstruction, while increasing key points will lead to decoding problems, such as adherence. In this paper, a symbol choice method with SDS and symbol features is proposed in order to obtain symbols with uniform distribution of more key points which will enhance the resolution and the decode rate.

3. PSM construction

3.1. Related conceptions about PSM

Before constructing a projected pattern, related concepts about M-array and De Bruijn are shown in Table 2. The projected pattern is built from a 1-D or 2-D matrix with the following mathematical properties. Let **P** be a matrix of dimension $r \times s$ where each element is taken from a alphabet of k symbols. If **P** has the window property, for instance, each different sub-matrix of $m \times n$ dimensions appears exactly once, then **P** is a M-array [12]. De Bruijn is a special M-array satisfying $r = m = 1$.

Table 3
Symbol design principles.

Principle	Contribution
Have more key points	Improve resolution of reconstruction
Accurate extraction of key points	Improve accurate of reconstruction
Easy decoding	For real time detection

A perfect submap is similar to a M-array except the latter one does not include all $m \times n$ sub-matrixes. This paper adopts perfect submap in coding pattern due to limitations in projector resolution and symbol size. Furthermore, choosing an appropriate window property helps the pattern accommodate surface occlusions and object shadows which may in turn increase the robustness of the system. In the coded pattern, more decoding points enhance the resolutions of reconstruction. In particular, extracting the points accurately in the deformed symbol increases the accuracy of reconstruction.

3.2. An optimal algorithm for constructing PSM

A larger PSM pattern size allows for higher resolution in the same measurement range. However, the pattern size of $m \times n$ is restricted to video graphics array (VGA) resolution and the symbol size ($r \times v$), which should be satisfied with M-array formula: Let $C = \{1, 2, \ldots, q\}$

$$m \times n \leqq q^{rv} \tag{1}$$

where q is the number of alphabets. The projected pattern is formed by q shape symbols primitive replace the alphabet $1, 2, \ldots, q$ in the M-array. For the projected pattern to appear clearly, there is one pixel white space between two symbols. Assume the symbols size is 7×7 pixels, and the projected pattern size is 1024×768 (VGA resolution). According to Eq. (1), to construct a 128×96 ($m \times n$) PSM pattern needs at least three symbols if the window dimension is 3×3 ($r \times v$). Due to the limitation of the VGA resolution and the sizes of the symbols, larger size of PSM pattern exceeds the system needs.

On the other hand, choosing an appropriate pattern window can enhance the robustness when dealing with occlusions and object shadows. A small window will improve the detection rate for distorted symbols which are damaged by sharp slope, specula reflection, and discontinuous surface. If one symbol in 2×2 window is damaged, the other three can be detected by neighboring windows. Compared with 3×3 window, damaged symbol in 2×2 window is much easier to repair. Therefore, it is best to choose the smallest window size possible.

Numerous pattern generation algorithms have been introduced previously [23,26,29,35,37] and can be classified into two categories: Hamming constraint or window unique constraint. Hamming constraint is equal to testing the Hamming distance in codeword space instead of the pattern array. Jia et al. [37] proposed a brute-force-like algorithm to check every step for window unique using piece growing algorithm to reduce the number of iteration. The method of Jia et al. is used to generate PSM in this paper. It randomly chooses pattern size ($m \times n$), number of symbols(q) and window size($r \times v$), which satisfies Eq. (1) to produce a distinct pattern size for different symbols with 2×2 window uniqueness.

4. PSM pattern design

Monochromatic coding methods are more robust against colored objects and can be segmented more easily compared with color coding methods. Most contributions focus on using different geometrical feature symbols to replace primitive in M-array or PSM [22–27]. The symbol design principle is shown in Table 3.

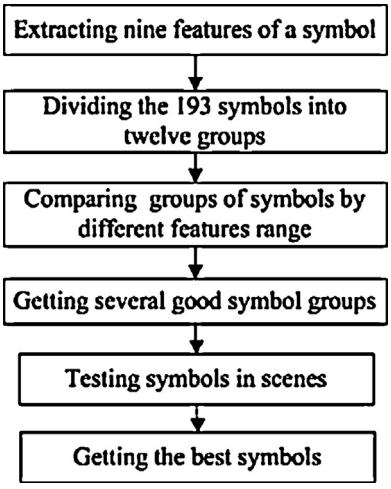


Fig. 1. Flowchart of the choice best symbols method.

Detection the center of intersection or its distortion is more accurate than circular-shaped features [25,26]. If symbols are designed with intersection will benefit to improve resolution of reconstruction. To improve the efficiency of the decoding algorithm, the symbol which has similar geometry feature can use the same decoding method. Similar geometric features, as an example, different symbols turn to show the same shape after being rotated by different angles. Following the symbol design principle, 193 special symbols were designed for best symbols test. The flowchart for acquiring the suitable symbols for the PSM pattern is shown in Fig. 1.

4.1. Symbol density spectrum (SDS)

Decoding deformed symbols with key points forms 3D reconstruction points. The key points are defined as a symbol's cross points (see Table 4). Symbols with a uniform distribution of a sufficient number of key points are searched in the 193 symbols. The ratio of key points is ambiguous in symbol design. A higher ratio of key points increase the resolution ratio; however, it may also cause distorted symbols and adherent key points. Moreover, the ratio of key points is does not often reflect the uniform distribution of the key points. Hence, this paper propose the symbol density spectrum (SDS) to represent key point distribution.

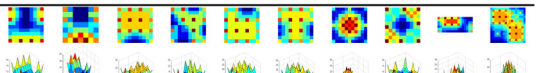
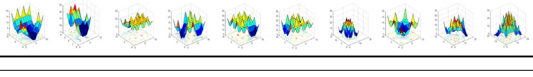
Suppose a symbol size is $i \times j$ and the point $f(x, y)$ is a key point which impacts the three layers around it. The δth layer $Layer_\delta(x_0, \delta)$ is defined as:

Let $x_0 \in \mathbb{R}^2$

$$Layer_\delta(x_0, \delta) = \{x | \delta \leqq \rho(x, x_0) \leqq \sqrt{2}\delta\} \tag{2}$$

where $\rho(x, x_0) \overset{def}{=} (\sum_{t=1}^{n} (x_t - x_{0_t})^2)^{1/2}$ is the distance between $x \in \mathbb{R}^2$ and $x_0 \in \mathbb{R}^2$, $\delta = 1, 2, 3$. For example, $Layer_1(x_0, 1)$ means point x_0 is a key point, x is in the first layer around the x_0. $Layer_1$ is a set

Table 4
Symbol density spectrum for key points.

Size:$i \times j$	10×10	7×7	10×10	10×10	10×10	10×10	10×10	10×10	7×16	17×17
Symbol_a										
SDS										
$Des(x, y)$										

[a]The key point is cross-point, which is marked by a red dot in a symbol.

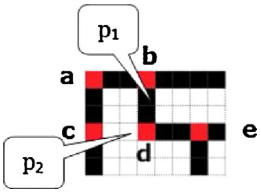


Fig. 2. Points p_1 and p_2 are influenced by different key points(marked with red) around them, $Des(x_{p_1}, y_{p_1})$ is sum of W_δ. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of points, which are the eight neighborhood points around the key point.

The SDS is defined as a set of $Des(x, y)$:

$$\mathrm{SDS} = \begin{pmatrix} Des(x_0, y_0) & Des(x_0, y_1) & \cdots & Des(x_0, y_{j-1}) \\ Des(x_1, y_0) & Des(x_1, y_1) & \cdots & Des(x_1, y_{j-1}) \\ \vdots & & \ddots & \\ Des(x_{i-1}, y_0) & Des(x_{i-1}, y_1) & \cdots & Des(x_{i-1}, y_{j-1}) \end{pmatrix} \tag{3}$$

where $Des(x, y)$ is the sum of corresponding W_δ in the symbol, that is:

$$Des(x, y) = \sum_{\delta=1}^{3} W_\delta \cdot Nkey_\delta \tag{4}$$

where $x \in [0, i-1]$, $y \in [0, j-1]$, $Nkey_\delta$ is the number of key points in $Layer_\delta(x, \delta)$. W_δ is the weight of reference layer, that is:

$$W_\delta = 10 - 2^\delta \tag{5}$$

where $\delta = 1, 2, 3$, and $W_0 = 16$. Example of $Des(x, y)$ in Fig. 2 are $Des(x_{p_1}, y_{p_1})$ and $Des(x_{p_2}, y_{p_2})$ which are the densities of points p_1 and p_2, respectively. The key points are marked with red. Point p_1 is in the layer of $Layer_1(x_b, 1)$, $Layer_2(x_d, 2)$, $Layer_3(x_a, 3)$, $Layer_3(x_e, 3)$, and $Layer_3(x_c, 3)$. Summing up these W_δ, $Des(x_{p_1}, y_{p_1})$ is 22. Similarly, point p_2 is in the layer of $Layer_1(x_d, 1)$, $Layer_2(x_c, 2)$, $Layer_3(x_a, 3)$, and $Layer_3(x_b, 3)$. The $Des(x_{p_2}, y_{p_2})$ is 22.

Some SDS for key points are shown in Table 4. Cooler colors in a SDS tends to have less key points around it. Key points uniform distribution reflects the density spectrum for symbols. As shown in Table 4, the three symbols(3rd, 5th,and 6th) presents uniform distribution for SDS compared with other symbols. Also, SDS is able to differentiate symbols that have the same geometry shape with different sizes, as shown in the 1st and 2nd symbol in Table 4.

4.2. Symbol feature extraction

For simplicity, several notations are defined as follows:

- $F\{\}$ If condition F is true, execute the operation in $\{\}$.
- $F\{[]:[]\}$: If condition F is true, execute the operation in the first [], otherwise execute the operation for the second [].
- $U = [0, i-1]$, $V = [0, j-1]$: Interval of U and V.
- $Count\{\}$: Record the number since the conditions were satisfied in $\{\}$.
- $dis(\mathbf{A} - \mathbf{B})$: The euclidean distance between **A** and **B**.

SDS reflects symbol's key points distribution. For comparative analysis of different symbols, symbol's features are extracted from SDS and symbols. Theoretically, key points rate *KeyNumRat* will improve the resolution, the definition of which is as follows:

$$keyNumRat = \frac{KeyN}{i \times j} \tag{6}$$

where $i \times j$ is the symbol size and *KeyN* is the number of key point.

KeyNumRat may not reflect the uniform distribution for the key points in a symbol. *PatKMas* are extracted from the SDS, which reflects key points dense distribution. *PatKMas* is the key point intensity of a symbol defined as follows:

$$PatKMas = F\{[1]:[NumM \times (\max_{x \in U, y \in V} Des(x, y) - W_0 + 1)]\} \tag{7}$$

where

$$NumM = Count\{Des(x, y) > W_1\}$$

Note:

$$x \in U, y \in V. \text{ when } x = 0, y = 0, \text{ initialize } NumM = 0,$$

$$F \text{ conditions} : NumM = 0$$

$\max_{x \in U, y \in V} Des(x, y)$ is the largest $Des(x, y)$ in a symbol. *NumM* is the sum of $Des(x, y)$ that should be greater than W_1 in a symbol. *NumM* counts the number of points in a symbol, which should satisfy that a point has at least one key points in its around three-layers. Higher *NumM* indicates that a symbol which has more key points in symbols point three-layer.

The larger *PatKMas* is, the denser the distribution of key points is and vice versa. Greater *PatKMas* could cause decoding problem such as key points adherence. *PatKMas* reflect high density of key point in a symbol, however, it still presents an accurate result when the density of key points is low and high in different parts of a symbol, respectively. Hence, a symbol whether has low density of key point, denoted by *NumWN*, is defined as follows:

$$NumWN = \frac{Count\{Des(x, y) = 0\}}{i \times j} \tag{8}$$

where $x \in U$, $y \in V$. Corresponding to *PatKMas*, *NumWN* reflects sparseness of a symbol. If the size and the number of key points of two symbols are the same, features of *NumWN* and *PatKMas* can judge the key points distribution. Symbols which have appropriate *PatKMas* and lower *NumWN* are suitable for PSM pattern, thus *KeyMean* combine the both features, is defined as follows:

$$KeyMean = F\left\{[NumWN]:[PatKMas + NumWN]\right\} \tag{9}$$

where $x \in U$, $y \in V$, F *conditions* : $NumM = 0$. Values of *PatKMas* and *NumWN* are normalized between $[-1, 1]$.

NoMeaRat and *JamARat* are another two features reflecting intensity and sparseness of symbol key points. Unlike the features mentioned above, the two features are extracted from the symbol image. If one random point for a symbol has no less than one key point in its eight neighborhoods, it has been measured, and vice versa. *NoMeaRat* is defined as follows:

$$NoMeaRat = 1 - \frac{Count\{Des(x, y) \geqslant 2W_1\} - KeyN}{i \times j - KeyN} \tag{10}$$

where $x \in U, y \in V$. A higher value of *NoMeaRat* indicates it has more area that does not measure in a symbol. *JamARat* reflects dense key points rate.

$$JamARat = \frac{Count\{Des(x, y) > 2W_1\}}{i \times j} \tag{11}$$

where $x \in U, y \in V$.

DisCtC is the distance between a symbol center c_s and SDS *CetMas*, the definition of which is as follows:

$$DisCtC = dis(CetMas - c_s) \tag{12}$$

where

$$CetMas = \left(\sum_{x=0}^{i-1} Des(x,y)/2, \sum_{y=0}^{j-1} Des(x,y)/2 \right)$$

A low *DisCtC* indicates that the SDS are based on central symmetry. *KStdDen* is standard deviation of SDS.

$$KStdDen = \frac{StdDen \times i \times j}{KeyN(\sum_{i=0}^{3} W_i \times 2^i)} \tag{13}$$

where

$$StdDen = \sqrt{\frac{1}{i \times j} \sum_{\substack{x \in U \\ y \in V}} \left(Des(x,y) - \frac{1}{i \times j} \sum_{\substack{x \in U \\ y \in V}} Des(x,y) \right)^2} \tag{14}$$

KeyN is the number of key points of a symbol. A low *KStdDen* indicates that the key points distribution tend to be very close to the mean.

The various features express different characteristics of symbols. Nine features have been extracted from symbol or SDS. The pattern is produced by several symbols to replace the primitive in PSM. Close symbols have similar features, which is beneficial to an even distribution in cloud points at 3D reconstruction. Similar symbols can adopt the same decoding algorithm improving decoding efficiency.

In order to assemble close symbols into one group, $PatSim_k$ is defined to measure similarity between symbols:

$$PatSim_k = Asc(NoMeaRat_k) + Asc(JamARat_k) + Desc(KeyNumRat_k) + Asc(KeyMean_k) \tag{15}$$

where $k \in [1, n]$, n is the total number of symbols, *Asc*( ) and *Desc*( ) are expressed as the symbol list in ascending and descending order, respectively. The kth symbol $PatSim_k$ is defined as the sum of order results for the four features: *NoMeaRat*, *JamARat*, *KeyNumRat*, and *KeyMean*. After being preprocessed, all the features are normalized between $[-1, 1]$. In addition, similar symbol presents similar SDS. For instance, the SDS of two symbols is the same after rotating different angles. Hence, according to the similarity of SDS and descending order of $PatSim_k$, all symbols are divided into 12 groups.

4.3. Analysis of group symbols feature

In order to simplify the feature contrast for the 193 symbols in the 12 groups, all nine features are computed using Eqs. (6)–(14). The features have been normalized between −1 and 1. Fig. 3 shows the result of the 12 groups' symbol features. Different colors in Fig. 3a, c, and e reflect the range of a feature for one group. The 12 groups have been divided into two sub-groups of the same size, as shown in Fig. 3a and c. Fig. 3e shows the comparison results for two groups in Fig. 3a and c. Fig. 3b is the central line corresponding to Fig. 3a, so as Fig. 3e and f.

KeyNumRat is the proportion of key points in a symbol. *PatKMas* reflects key points dense distribution extracted from SDS. Higher *KeyNumRat* and *PatKMas* contribute to higher resolution of 3D reconstruction also in addition to increasing decoding problems such as adherence. After being normalized, smaller features are better except for *KeyNumRat* and *PatKMas*.

As shown in Fig. 3a, group 4 is the best features compared with the other five groups. Group 8 is the second. Most of features values of the two groups are less than zero. Compared with group 4, values of features in group 8 fluctuate widely. Actually, feature value is steadily conducive to 3D detection. Compared with other groups shown in Fig. 3a, group 3 and group 12 have poor performance, due to the wider ranges of their feature values. For example, as shown in Fig. 3a, group 12 has wider and higher values for features of *KeyNumRat* and *PatKMas*. This indicates the key points distribution is too dense in group 12. Select group 12 symbols are shown in Table 5.

Table 5
List comparative results for symbol groups.

Group number	Symbol size	Sx_0	Sx_1	Sx_2	Sx_3	Sx_4	Sx_5	Sx_6	Sx_7	Sx_8
4	10×10									
6_1										
6_2										
6_3										
12	7×7									

The rest of sub-groups are shown in Fig. 3c and d. By contrast, group 5 and group 6 are better than others. As shown in Fig. 3(e) and 3(f), four groups(4, 5, 6, and 8) are picked from Fig. 3a and c. The four groups have better results compared with other groups as shown in Fig. 3a and c. Among the four groups, group 8 fluctuates more widely than the others, which means features are unsteady in the group. Group 5 and group 8 have higher values of *NoMeaRat* than other groups. As shown in (10), a higher value of *NoMeaRat* indicates the pattern would lose surface detail in detection. As shown in Fig. 3e and f, group 4 and group 6 have similar results. Table 5 lists the symbols used in group 4 and group 6. Group 6_1, 6_2, and 6_3 are the subdivisions of group 6, according to the same shape after rotation.

4.4. Choice of the best symbols

Group 4 and group 6 have better symbol features compared with other groups which have 36 symbols in total. A few suitable symbols are sought to generate PSM pattern. Thus, symbols also are tested in a real scene for evaluation. In experiment, patterns are generated by the symbols that are chosen randomly in every group, which are projected to the scene and are decoded to choose best groups.

The experiments for testing symbols in the scene is described in detail as follows.

Step 1: According to feature $PatSim_k$, 193 symbols are divided into 12 groups. For symbols shown clearly in the scene, the pattern is formed by white symbols and black background, as shown in Fig. 4.
Step 2: Projecting patterns onto a different scene are respective, such as on irregular or color surfaces, occlusions, and object shadows, as shown in Fig. 2.
Step 3: Decode the distorted pattern and analyze the error. The symbols in circles shows the adherence of key points in Fig. 4a. Sparse key points fails to measure the detail of the surface, as shown in Fig. 4(b). Symbols become incomplete due to shadow, specular reflection and high curvatures, as shown in Fig. 4c, d, and e, respectively.
Step 4: Order the groups by symbol damage rate.

Select testing results for corresponding symbol pattern are presented in Table 6. Typical difficulties on symbol decoding (shadow,

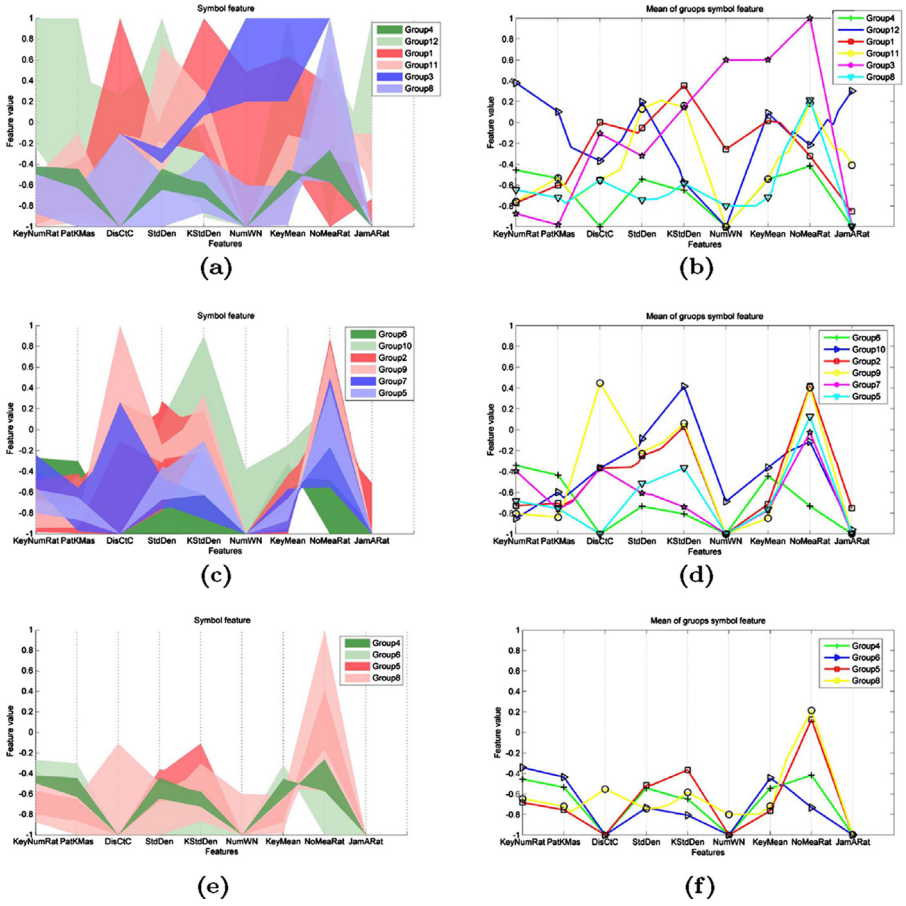


Fig. 3. Symbol feature contrast with 12 different groups, (a) is the random six groups, (c) is the rest of groups. (e) Pick better four groups symbols contrast in (a) and (c). Different colors in the picture shows one group features range. Apart from the feature of *KeyNum* and *PatKMas*, smaller features are better for the symbol. (b) is mean line corresponded to (a), so as (e) and (f).

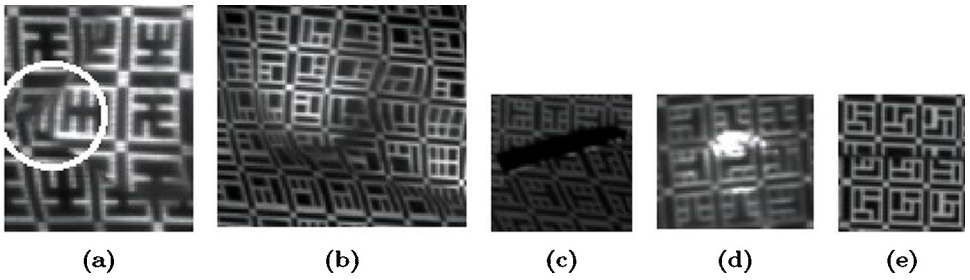


Fig. 4. Symbol distortion causes some bad consequences. (a) Key points adhere (the symbol in circle), (b) sparse key points, (c) symbol damaged by shadows, (d) specular reflection, (e) high curvatures.

illumination and rapid surface change) cannot be resolved by our method. However, the aim of the current study is to enhance resolution for spatial coding and to decrease symbol damage rate when pattern is deformed.

As shown in Table 6, *SymDamRat* is damage rate of symbols defined as:

$$SymDamRat = \frac{NumDam}{NumSyPat}$$

Table 6
Group 4 and Group 6 corresponding pattern testing in scene.

Surface	*SymDamRat*				*NumKPat*[a]			
	Group 4	Group 6_1	Group 6_2	Group 6_3	Group 4	Group 6_1	Group 6_2	Group 6_3
Plan	1.172%	1.096%	1.153%	1.190%	72,634	79,095	79,049	74,378
Stairs	1.493%	1.455%	1.587%	1.398%	72,399	78,810	78,704	74,224
Cylinder	2.646%	1.512%	1.946%	2.494%	71,506	78,763	78,417	73,398
Irregular surface	14.758%	5.159%	9.921%	12.283%	62,640	75,850	72,045	66,042

[a] *SymDamRat* is symbol damage rate. *NumKPat* is the number of key points for a pattern in theoretical decoding.

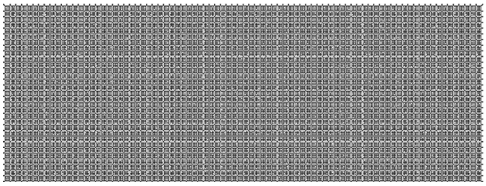


Fig. 5. Partial symbol pattern of 63 × 84 PSM with 2 × 2 window property.

where *NumDam* is the sum of damage symbols in a pattern. *NumDam* includes the symbols are damaged by key points adhering, shadows, reflecting light, and severe deformations caused by high curvatures. *NumSyPat* and *NumKPat* are the number of symbols and key points, respectively. A higher *SymDamRat* mean the pattern has higher rate of failure in decoding. Meanwhile the higher the *NumKPat* is, the higher the resolution will be.

As shown in Table 6, patterns have been tested on plan, stairs, cylinder, and irregular surfaces. The physical location of the CCD camera, projector and object, and focal distance are constant in pattern testing. As symbol decoding methods are different from group to group, it is difficult to compare different decoding methods. In order to simplify symbol comparing, suppose that all the symbols can be decoded if it has not been destroyed.

The irregular surface has a higher *SymDamRat* compared with other test surfaces, as shown in Table 6, as surface fluctuation causes an increase of key points adhesion, damaged symbols and shadow. Group 6_1 is better than other groups especially in the irregular surface. The paper adopted group 6_1 for producing the PSM pattern in a 3D scanner. Only a part of PSM symbol pattern is shown in Fig. 5. The pattern size is 63 × 84 with 2 × 2 window property. As shown in Fig. 5, symbols are connected with lines, which forms the pattern as a network. The line make a symbol in four windows simultaneously. If there are errors or fails when identifying the symbols, they can be corrected by the adjacent symbols for the window uniqueness.

5. Experimental results and analysis

5.1. Experimental conditions

The system setup is a 3D measurement system, which is coded by one-shot monochromatic PSM. This system is designed to scan the surface of objects. The experiment test the monochromatic symbol pattern by depth accuracy, spatial density, and whether suitable for colorful surface. The inspection system consists of four fundamental parts: commercial projectors (Epson EMP-821 Series LCD Projectors, with 1024 × 768 pixel size, 20 Hz project speed), a high resolution commercial monochromatic CCD camera (Dalsa monochromatic Falcon Series: DS-24-02M30, with 1920 × 1080 resolution), CCTV Lens (Kowa LM16HC or LM25HC, Focal Length 35.0 mm) and a computer (Intel core i5, Memory: DDR3 8 GB 1600 Mhz).

5.2. Surface reconstruction and error analysis

In the experiment, the symbol pattern (63 × 84) described in Section 4.4 (Fig. 5) is projected to the scene. The picture is captured by a camera to save the projected light which is then modulated by the object(Fig. 8b). A decoding method then enhances decoding rate using a multi-threshold and the window vote algorithm. The reconstruction process is described in Figs. 6 and 8. As shown in Fig. 8, the pattern is projected onto a colorful book cover. The numbers are the location of the corresponding decoded symbols in the PSM pattern, as illustrated in Fig. 8d. The number of −1 shows the decoding error for the corresponding symbol.

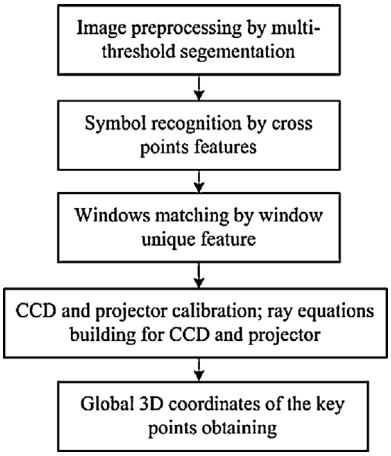


Fig. 6. Flowchart of the reconstruction process.

Select results for each step are presented in Table 7. The accuracy of symbol recognition is higher than 94% at the step of segmented regions, as shown in Table 7. The best detected rate is 99.7% for the flat surface. Irregular surfaces have a lower detection rate compared with gently rolling surfaces.

Symbols are usually cannot be recognized in the following cases. Firstly, the uneven of illumination intensity in a symbol would cause the failure in symbol segmentation. Although the multi-threshold decoding method can deal with uneven illumination for the whole distorted pattern, it would fail in decoding for one symbol. As shown in Fig. 8c and d, symbols failed to recognize the titles on the book cover due to the uneven illumination in one symbol. Secondly, symbols are "broken" by object shadows, specular reflection, and high curvatures. Thirdly, the orientation of the surface changes rapidly from the view of the camera sighting, such as mouth edges(see Fig. 7i). Fourthly, key points detection is invalid for serious defocus blur. As shown in Fig. 7b, a part of plant is not detected due to serious defocus blur. Lastly, as the colors of the symbol and its background are similar, it is difficult to separate the symbol from the background. As shown in Fig. 9a and b, the color of the road in the image is too close to black; therefore, it is failed to identify symbols.

The error detection rate is ca. 0.06% for the ruled surface. The value of the error detection rate increases when the test is performed on the surface of irregular and colorful objects.

Depth acquisition experiments were conducted for different objects, including a flat surface, stairs, a dish, and plaster models of mouth and nose. The results of object reconstruction are shown in Fig. 7.

5.3. Depth accuracy

For most objects, it is not easy to estimate the absolute accuracy of the measurement due to the lack of the ground-truth data for detecting surfaces. This paper tests the accuracy of the system in an indirect way. A flat surface is acquired at three different distances from the camera. The depth errors at each point are calculated relative to the fitted plane. The accuracy performance is evaluated based on the standard deviation of these errors. It is found that the distance between the camera and the flat surface has little effect on the relative error standard deviation. The surface plane is set at distances of 143, 162, and 173 mm from the camera with standard deviations of 0.5324, 0.5811, and 0.6032 mm, respectively. The errors are mainly caused by key points detection and projector calibration. The problems of corner detection are not the focus of this paper. The precision is suitable for most intended applications.

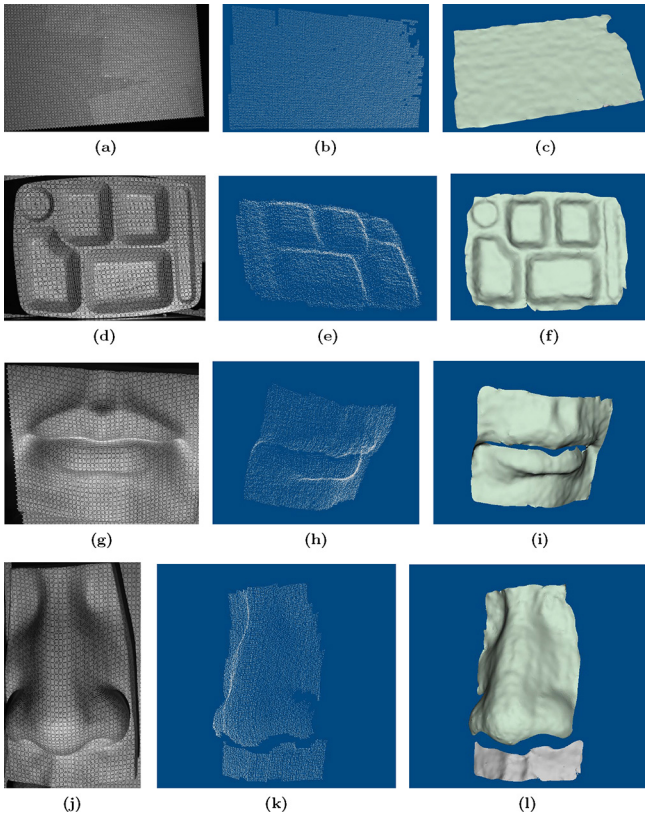


Fig. 7. Objects reconstruction. (a) Flat surface, (b) cloud points of the plan, (c) reconstruction for the plan, (d) dish, (e) cloud points of the dish, (f) reconstruction for the dish, (g) plaster models of mouth, (h) cloud points of the mouth, (i) reconstruction for the mouth, (j) plaster models of nose, (k) cloud points of the nose, (l) reconstruction for the nose.

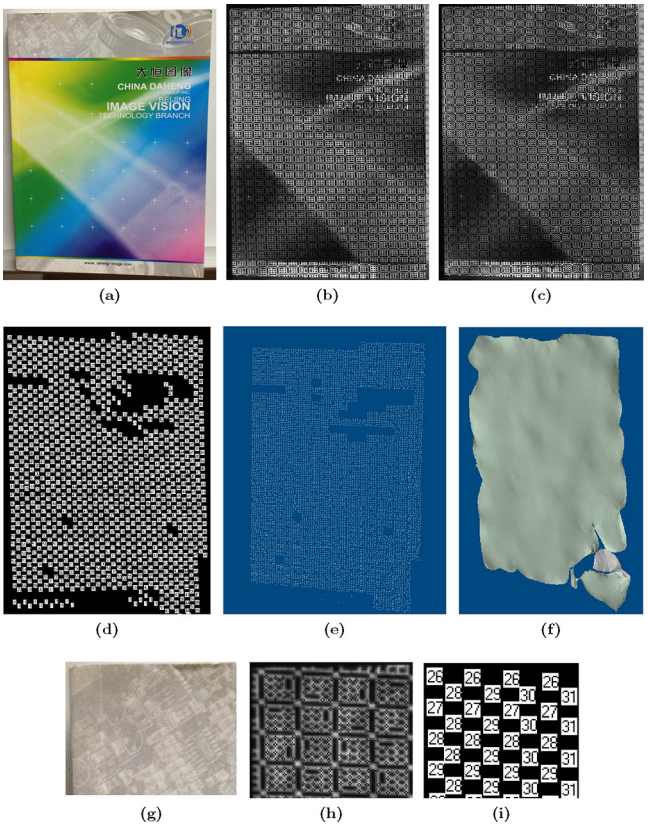


Fig. 8. Decoding process. (a) Target object, (b) captured scene with PSM coding method, (c) decoding symbols with key points. Key points are marked with a cross, (d) symbol location in the PSM pattern, (e) cloud points for the plan, (f) reconstructed shape. (g) Detail for (a), (h) detail for (c), (i) detail for (d).

Furthermore, it should be pointed out that with higher resolution camera and projector, and sub-pixel detection of the key points, the accuracy can liked be improved.

5.4. Testing on colorful surface and discussion

As shown in Figs. 8 and 9, the experimental results demonstrate the robustness of the symbol-based pattern against colorful objects. The system fail to recognize part of symbols as shown in Figs. 8 and 9. The reasons have been listed in Section 5.2. The decoding method performs well in the case of uneven brightness images (Fig. 8c and h) and image noise (Fig. 9c). However, the error detection rate increases for colorful surfaces.

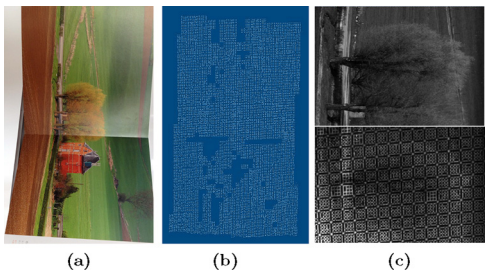


Fig. 9. (a) Colorful surface, (b) cloud points of the colorful surface, (c) detail for decoding. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.5. Spatial density

Table 8 lists references for one-shot monochromatic coding pattern, M-array, De Bruijn, and other related method (Table 2). Primitives are replaced by geometrical shapes such as disks, rings, segments, cuneiforms, squares, and special design symbols. The proposed pattern has most key points, which will contribute to improving the resolution of reconstruction. Ref. [26] utilizes the largest pattern size and the minimum symbol size to reconstruct 3D surface. However, for each symbol, there is only one key point. Moreover, in Ref. [26] the patterns are subperfect submaps, which do not ensure the uniqueness of the window, and therefore are

Table 7
Results of several processes in the recognition.

Recognition data evaluated	Experiments				
	Flat surface	Cylinder	Stairs	Dish	Mouth
Segmented regions of symbols	5292	5292	3974	1147	1419
Detected symbols	5277	5212	3911	1097	1346
Measured key points	79,682	78,701	59,056	16,465	20,219
Error detection rate	0%	0.057%	0.050%	0.054%	0.081%
Percentage of recognized	99.71%	97.92%	97.91%	96.16%	94.0%

Table 8
One-shot monochromatic M-array(or related concepts) pattern.

Reference	[22]	[23]	[24]	[25,26]	[27]	[c]
1.[a]	27 × 127	27 × 29	79 × 59	100 × 150 [26][b]	54 × 36	63 × 84
2.	?	3 × 3	2 × 2	3 × 3	3 × 3	2 × 2
3.	5	3	10	4	5	9
4.	?	?	11 × 11	8 × 8	18 × 12	10 × 10
5.	1	1	6	1	4 or 6	15 or 16
6.	3,429	783	27,966	15,000	11,232	79,972

[a] 1. Pattern size, 2. Sub-window size, 3. Number of primitives, 4. The mean area per symbol, 5. Key points for one symbol has, 6. Number of key points for decoding pattern (at best case).
[b] Refs. [25,26] design difference pattern sizes testing in scene, size of 100 × 150 is the biggest one. Refs. [25,26] has eight symbols, use four symbols to produce a pattern.
[c] Proposed pattern.

restrained by epipolar lines. Similar to Ref. [24], this paper designs special symbols as well, which increases the key point's number by 284%.

6. Conclusion

In this paper, we have proposed new monochromatic symbols to form the projection pattern, which is extracted by comparing symbol features and their SDS. The proposed SDS and symbol's features are used to choose proper symbols. In the experiment, the one-shot monochromatic symbols pattern gains almost 78,000 points for reconstruction and has higher resolution compared with other monochromatic spatial coding strategies.

The primitive in PSM is replaced by special designed symbols to form the projection pattern. The symbols tested in the scene have several advantages. First, the designed symbols have more key points for decoding, which improves the resolution of reconstruction. Second, adopted symbols have the smallest symbol damage rate. As a result, the decoding rate is effectively enhanced. In addition, symbols have similar geometric features, which improve the efficiency of the decoding. Finally, the pattern is coding with monochromatic symbols. Therefore, it is less sensitive to color surface when compared to color coding. Quantitative and qualitative comparisons in the experimental results confirm that one-shot monochromatic symbol pattern is robust to colorful objects, and produce 3D reconstructions with high resolution.

Acknowledgements

The work is supported by National Natural Science Foundation of China (No. 61303095), Scientific Research Innovation Foundation of Shanghai Municipal Education Commission (No. 14YZ018) and also supported by Science and Technology Plan of Jiaxing (No. 2013AY11013).

References

[1] J. Salvi, S. Fernandez, T. Pribanic, X. Llado, A state of the art in structured light patterns for surface profilometry, Pattern Recognit. 43 (8) (2010) 2666–2680.
[2] R. Sagawa, Y. Ota, Y. Yagi, R. Furukawa, N. Asada, H. Kawasaki, Dense 3D reconstruction method using a single pattern for fast moving object, in: 2009 IEEE 12th International Conference on Computer Vision, 2009, pp. 1779–1786.
[3] S. Zhang, Recent progresses on real-time 3D shape measurement using digital fringe projection techniques, Opt. Lasers Eng. 48 (2) (2010) 149–158.
[4] S. Zhang, S.-T. Yau, High-speed three-dimensional shape measurement system using a modified two-plus-one phase-shifting algorithm, Opt. Eng. 46 (11) (2007) 113603.
[5] J. Xu, B. Gao, J. Han, J. Zhao, S. Liu, Q. Yi, Z. Zhao, H. Yin, K. Chen, Realtime 3D profile measurement by using the composite pattern based on the binary stripe pattern, Opt. Laser Technol. 44 (3) (2012) 587–593.
[6] Y. Fu, G. Jiang, F. Chen, A novel Fourier transform profilometry based on dual-frequency grating, Optik 123 (10) (2012) 863–869.
[7] Y. Fu, Z. Wang, J. Yang, J. Wang, G. Jiang, Three-dimensional profile measurement of the blade based on multi-value coding, Optik 125 (11) (2014) 2592–2596.
[8] Z. Zhang, D.P. Towers, C.E. Towers, Snapshot color fringe projection for absolute 3D metrology of video sequences, Appl. Opt. 49 (31) (2010) 5947–5953.
[9] N. Karpinsky, S. Zhang, Composite phase-shifting algorithm for three-dimensional shape compression, Opt. Eng. 49 (6) (2010) 063604.
[10] Z.H. Zhang, Review of single-shot 3D shape measurement by phase calculation-based fringe projection techniques, Opt. Lasers Eng. 50 (8) (2012) 1097–1106.
[11] C. Guan, L.G. Hassebrook, D.L. Lau, Composite structured light pattern for three-dimensional video, Opt. Express 11 (5) (2003) 406–417.
[12] J. Salvi, J. Pages, J. Batlle, Pattern codification strategies in structured light systems, Pattern Recognit. 37 (4) (2004) 827–849.
[13] L. Zhang, B. Curless, S.M. Seitz, Rapid shape acquisition using color structured light and multi-pass dynamic programming, in: Proceedings. First International Symposium on 3D Data Processing Visualization and Transmission, 2002, IEEE, 2002, pp. 24–36.
[14] X. Zhang, Y. Li, L. Zhu, Discontinuity-preserving decoding of one-shot shape acquisition using regularized color, Opt. Lasers Eng. 50 (10) (2012) 1416–1422.
[15] J. Pages, J. Salvi, J. Forest, A new optimised de Bruijn coding strategy for structured light patterns, in: ICPR 2004. Proceedings of the 17th International Conference on Pattern Recognition, Vol. 4, IEEE, 2004, pp. 284–287.
[16] J. Pages, C. Collewet, F. Chaumette, J. Salvi, An approach to visual servoing based on coded light, in: ICRA 2006. Proceedings 2006 IEEE International Conference on Robotics and Automation, 2006, pp. 4118–4123.
[17] A. Adan, F. Molina, A.S. Vzquez, L. Morena, 3D feature tracking using a dynamic structured light system, in: Proceedings. The 2nd Canadian Conference on Computer and Robot Vision, 2005, IEEE, 2005, pp. 168–175.
[18] A. Adan, A.S. Vzquez, C. Cerrada, S. Salamanca, Moving surface extraction based on unordered hexagonal perfect submaps projection: applications to 3d feature tracking, Image Vis. Comput. 27 (8) (2009) 1083–1096.
[19] H. Chen, S. Ma, Feature points matching for face reconstruction based on the window unique property of pseudo-random coded image, Mechatronics 22 (6) (2012) 688–695.
[20] D. Desjardins, P. Payeur, Dense stereo range sensing with marching pseudo-random patterns, in: CRV'07. Fourth Canadian Conference on Computer and Robot Vision, 2007, IEEE, 2007, pp. 216–226.
[21] Z. Song, C.-K.R. Chung, Determining both surface position and orientation in structured-light-based sensing, IEEE Trans. Pattern Anal. Mach. Intell. 32 (10) (2010) 1770–1780.
[22] P. Griffin, L. Narasimhan, S. Yee, Generation of uniquely encoded light patterns for range data acquisition, Pattern Recognit. 25 (6) (1992) 609–616.
[23] C. Albitar, P. Graebling, C. Doignon, Robust structured light coding for 3D reconstruction, in: ICCV 2007. IEEE 11th International Conference on Computer Vision, 2007, 2007, pp. 1–6.
[24] X. Jia, Z. Zhang, F. Cao, D. Zeng, Model and error analysis for coded structured light measurement system, Opt. Eng. 49 (12) (2010) 123603.
[25] X. Maurice, P. Graebling, C. Doignon, Epipolar based structured light pattern design for 3-D reconstruction of moving surfaces, in: IEEE International Conference on Robotics and Automation (ICRA), 2011, 2011, pp. 5301–5308.
[26] X. Maurice, P. Graebling, C. Doignon, A pattern framework driven by the hamming distance for structured light-based reconstruction with a single image, in: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2011, 2011, pp. 2497–2504.
[27] M.L.L. Reiss, A.M.G. Tommaselli, A low-cost 3D reconstruction system using a single-shot projection of a pattern matrix, Photogramm. Rec. 26 (133) (2011) 91–110.
[28] S. Fernandez, J. Salvi, One-shot absolute pattern for dense reconstruction using Debruijn coding and windowed Fourier transform, Opt. Commun. 291 (2013) 70–78.
[29] K. Claes, Structured light adapted to control a robot arm, Katholieke Universiteit Leuven, 2008 (Ph.D. thesis).
[30] T.P. Koninckx, L.V. Gool, Real-time range acquisition by adaptive structured light, IEEE Trans. Pattern Anal. Mach. Intell. 28 (3) (2006) 432–445.

[31] P. Vuylsteke, A. Oosterlinck, Range image acquisition with a single binary-encoded light pattern, IEEE Trans. Pattern Anal. Mach. Intell. 12 (2) (1990) 148–164.
[32] Q. Yu, X. Jia, J. Tao, Y. Zhao, An encoded mini-grid structured light pattern for dynamic scenes, Adv. Intell. Comput. (Springer) (2005) 126–135.
[33] M.K. Siu, The combinatorics of binary arrays, J. Stat. Plan. Inference 62 (1) (1997) 103–113.
[34] T. Etzion, Constructions for perfect maps and pseudorandom arrays, IEEE Trans. Inf. Theory 34 (5) (1988) 1308–1316.
[35] R. Morano, C. Ozturk, R. Conn, S. Dubin, S. Zietz, J. Nissanov, Structured light using pseudorandom codes, IEEE Trans. Pattern Anal. Mach. Intell. 20 (3) (1998) 322–327.
[36] F. MacWilliams, N.J.A. Sloane, Pseudo-random sequences and arrays, Proc. IEEE 64 (12) (1976) 1715–1729.
[37] X.-J. Jia, Z.-J. Zhang, Q.-C. Yu, Construction for m-arrays and application in structured light, J. Shanghai Univ. (Engl. Ed.) 15 (1) (2011) 63–68.